KAIST

2026 General Relativity

Homework 2

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#1

The field strength tensor of $\mathbf{E} = (E_x, E_y, E_z), \mathbf{B} = (B_x, B_y, B_z)$ is

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} \tag{1}$$

(a)

Transformation to a frame rotated by θ about the y-axis is given by

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & 0 & -\sin\theta \\ 0 & 0 & 1 & 0 \\ 0 & \sin\theta & 0 & \cos\theta \end{pmatrix} \tag{2}$$

The field strength tensor in the rotated frame is given by

$$F'_{\mu\nu} = \Lambda_{\mu}{}^{\alpha}\Lambda_{\nu}{}^{\beta}F_{\alpha\beta} \tag{3}$$

This can be expressed by matrix multiplication as

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & 0 & -\sin\theta \\ 0 & 0 & 1 & 0 \\ 0 & \sin\theta & 0 & \cos\theta \end{pmatrix} \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & 0 & \sin\theta \\ 0 & 0 & 1 & 0 \\ 0 & -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -E_x\cos\theta + E_z\sin\theta & -E_y & -E_z\cos\theta - E_x\sin\theta \\ E_x\cos\theta - E_z\sin\theta & 0 & B_z\cos\theta + B_x\sin\theta & -B_y \\ E_y & -B_z\cos\theta - B_x\sin\theta & 0 & B_x\cos\theta - B_z\sin\theta \\ E_z\cos\theta + E_x\sin\theta & B_y & -B_x\cos\theta + B_z\sin\theta & 0 \end{pmatrix} \tag{4}$$

Hence, the transformed electric and magnetic fields are

$$\mathbf{E}' = (E_x\cos\theta - E_z\sin\theta, E_y, E_z\cos\theta + E_x\sin\theta), \tag{5}$$

$$\mathbf{B}' = (B_x\cos\theta - B_z\sin\theta, B_y, B_z\cos\theta + B_x\sin\theta) \tag{6}$$

(b)

Transformation to a frame boosted by β along the z-axis is given by

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & 0 & 0 & -\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \tag{7}$$

As same as (3, 4),

$$\begin{pmatrix} \gamma & 0 & 0 & \beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \beta\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{pmatrix} \begin{pmatrix} \gamma & 0 & 0 & \beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \beta\gamma & 0 & 0 & \gamma \end{pmatrix}$$
$$= \begin{pmatrix} 0 & -\gamma E_x + \beta\gamma B_y & -\gamma E_y - \beta\gamma B_x & -E_z \\ \gamma E_x - \beta\gamma B_y & 0 & B_z & -\gamma B_y + \beta\gamma E_x \\ \gamma E_y + \beta\gamma B_x & -B_z & 0 & \gamma B_x + \beta\gamma E_y \\ E_z & \gamma B_y - \beta\gamma E_x & -\gamma B_x - \beta\gamma E_y & 0 \end{pmatrix} \tag{8}$$

Hence, the transformed electric and magnetic fields are

$$\mathbf{E}' = (\gamma E_x - \beta\gamma B_y, \gamma E_y + \beta\gamma B_x, E_z), \tag{9}$$
$$\mathbf{B}' = (\gamma B_x + \beta\gamma E_y, \gamma B_y - \beta\gamma E_x, B_z) \tag{10}$$

#2

Energy-momentum tensor is Noether current associated with spacetime translation symmetry.

$$T^{\mu\nu} = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_\alpha)}\partial^\nu \phi_\alpha + \eta^{\mu\nu}\mathcal{L} \tag{11}$$

(a)

For the source-free Maxwell's electromagnetism, $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, and $\phi_\alpha = A_\alpha$. Hence, we have

$$-\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\alpha)}\partial^\nu A_\alpha + \eta^{\mu\nu}\mathcal{L} = F^{\mu\alpha}\partial^\nu A_\alpha + \eta^{\mu\nu}\mathcal{L} \tag{12}$$

$$= F^{\mu\alpha}(\partial^\nu A_\alpha - \partial_\alpha A^\nu) + F^{\mu\alpha}\partial_\alpha A^\nu + \eta^{\mu\nu}\mathcal{L} \tag{13}$$

$$= F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + F^{\mu\alpha}\partial_\alpha A^\nu \tag{14}$$

Using the equation of motion $\partial_\alpha F^{\mu\alpha} = 0$, The last term is

$$F^{\mu\alpha}\partial_\alpha A^\nu = \partial_\alpha(F^{\mu\alpha}A^\nu) - (\partial_\alpha F^{\mu\alpha})A^\nu = \partial_\alpha(F^{\mu\alpha}A^\nu) \tag{15}$$

And this term is conserved standing alone,

$$\partial_\mu\partial_\alpha(F^{\mu\alpha}A^\nu) = -\partial_\alpha\partial_\mu(F^{\alpha\mu}A^\nu) = -\partial_\mu\partial_\alpha(F^{\mu\alpha}A^\nu) \tag{16}$$

$$\implies \partial_\mu\partial_\alpha(F^{\mu\alpha}A^\nu) = 0 \tag{17}$$

Hence, we get the symmetric and gauge-invariant energy-momentum tensor for Maxwell's electromagnetism as

$$T^{\mu\nu} = F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} \tag{18}$$

To express it in terms of $\mathbf{E}$ and $\mathbf{B}$, separate the time component and space block. The time component is

$$T^{00} = F^{0\alpha}F^{0}{}_{\alpha} - \frac{1}{4}\eta^{00}F_{\alpha\beta}F^{\alpha\beta} \tag{19}$$

$$= E_x^2 + E_y^2 + E_z^2 + \frac{1}{4}(2B_x^2 + 2B_y^2 + 2B_z^2 - 2E_x^2 - 2E_y^2 - 2E_z^2) \tag{20}$$

$$= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) \tag{21}$$

For the space block, use the expression $F^{ij} = \varepsilon^{ijk}B_k$.

$$T^{ij} = F^{i0}F^{j}{}_{0} + F^{ik}F^{j}{}_{k} - \frac{1}{4}\eta^{ij}F_{\alpha\beta}F^{\alpha\beta} \tag{22}$$

$$= -E_iE_j + \varepsilon^{ikl}\varepsilon^{jkm}B_lB_m - \frac{1}{2}\delta_{ij}(\mathbf{B}^2 - \mathbf{E}^2) \tag{23}$$

$$= -E_iE_j + (\delta_{ij}\delta_{lm} - \delta_{im}\delta_{jl})B_lB_m - \frac{1}{2}\delta_{ij}(\mathbf{B}^2 - \mathbf{E}^2) \tag{24}$$

$$= -E_iE_j - B_iB_j + \frac{1}{2}\delta_{ij}(\mathbf{E}^2 + \mathbf{B}^2) \tag{25}$$

For the mixed component is

$$T^{j0} = F^{j\alpha}F^{0}{}_{\alpha} - \frac{1}{4}\eta^{j0}F_{\alpha\beta}F^{\alpha\beta} \tag{26}$$

$$= F^{jk}F^{0}{}_{k} = \varepsilon^{jkl}B_l E_k \tag{27}$$

$$= (\mathbf{E}\times\mathbf{B})_j \tag{28}$$

($F^{0}{}_{0} = 0$, $\eta^{j0} = 0$, and $F^{0k} = +E_k$ are used.)

Therefore,

$$T^{\mu\nu} = \begin{pmatrix} \frac{1}{2}(\mathbf{E}^2+\mathbf{B}^2) & (\mathbf{E}\times\mathbf{B})^\top \\ \mathbf{E}\times\mathbf{B} & -\mathbf{E}\mathbf{E}^\top - \mathbf{B}\mathbf{B}^\top + \frac{1}{2}(\mathbf{E}^2+\mathbf{B}^2)I \end{pmatrix} \tag{29}$$

For the scalar field theory, $\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)$, and $\phi_\alpha = \phi$, The energy momentum tensor is

$$T^{\mu\nu} = \partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}\left(\frac{1}{2}\eta^{\alpha\beta}\partial_\alpha\phi\partial_\beta\phi + V(\phi)\right) \tag{30}$$

The time component is

$$T^{00} = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi) \tag{31}$$

And the space block is

$$T^{ij} = \partial^i\phi\partial^j\phi - \delta^{ij}\left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi)\right) \tag{32}$$

the mixed component is

$$T^{j0} = -\dot{\phi}\partial_j\phi \tag{33}$$

Therefore,

$$T^{\mu\nu} = \begin{pmatrix} \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi) & -(\dot{\phi}\nabla\phi)^\top \\ -(\dot{\phi}\nabla\phi) & \nabla\phi\nabla\phi^\top - \left(-\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi)\right)I \end{pmatrix} \tag{34}$$

(b)

For the source-free Maxwell's electromagnetism, the equation of motion is $\partial_\mu F^{\mu\nu} = 0$. Hence, the conservation law $\partial_\mu T^{\mu\nu} = 0$ is

$$\partial_\mu T^{\mu\nu} = \partial_\mu F^{\mu\alpha}F^{\nu}{}_{\alpha} + F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}\partial_\mu(F_{\alpha\beta}F^{\alpha\beta}) \tag{35}$$

$$= F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}\partial_\mu(F_{\alpha\beta}F^{\alpha\beta}) \tag{36}$$

$$= F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} - \frac{1}{2}\eta^{\mu\nu}F^{\alpha\beta}\partial_\mu F_{\alpha\beta} \tag{37}$$

$$= F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} + \frac{1}{2}\eta^{\mu\nu}F^{\alpha\beta}(\partial_\alpha F_{\beta\mu} + \partial_\beta F_{\mu\alpha}) \tag{38}$$

$$= F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} + \eta^{\mu\nu}F^{\alpha\beta}\partial_\alpha F_{\beta\mu} \tag{39}$$

$$= F^{\mu\alpha}\partial_\mu F^{\nu}{}_{\alpha} - F^{\alpha\beta}\partial_\alpha F^{\nu}{}_{\beta} \tag{40}$$

$$= 0 \tag{41}$$

Where the Bianchi identity $\partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} + \partial_\alpha F_{\mu\nu} = 0$ is used in the fourth line.

As same as above, for the scalar field theory, the equation of motion is $\partial_\mu \partial^\mu \phi - V'(\phi) = 0$. Hence, the conservation law $\partial_\mu T^{\mu\nu} = 0$ is

$$\partial_\mu T^{\mu\nu} = \partial_\mu \partial^\mu \phi \partial^\nu \phi + \partial^\mu \phi \partial_\mu \partial^\nu \phi - \eta^{\mu\nu} \left(\frac{1}{2} \partial_\mu (\partial^\alpha \phi \partial_\alpha \phi) + V'(\phi) \partial_\mu \phi \right) \tag{42}$$

$$= \partial_\mu \partial^\mu \phi \partial^\nu \phi + \partial^\mu \phi \partial_\mu \partial^\nu \phi - \eta^{\mu\nu} \left(\partial^\alpha \phi \partial_\alpha \partial_\mu \phi + V'(\phi) \partial_\mu \phi \right) \tag{43}$$

$$= (\partial_\mu \partial^\mu \phi - V'(\phi)) \partial^\nu \phi + \partial^\mu \phi \partial_\mu \partial^\nu \phi - \eta^{\mu\nu} \partial^\alpha \phi \partial_\alpha \partial_\mu \phi \tag{44}$$

$$= 0 \tag{45}$$

#3

(a)

Let V be a n-dimensional vector space, and let g be a metric on V (i.e., a symmetric, non-degenerate, bilinear form).

For $n = 1$, non-degeneracy condition implies $g(e, e) \neq 0$ for any non-zero $e \in V$. Hence, we can choose a basis $\{e\}$ such that $g(e, e) = \pm 1$ by picking a non-zero vector and normalizing it.

Suppose that any k-dimensional vector space has a basis $\{e_1, \ldots, e_k\}$ such that $g(e_i, e_j) = \pm\delta_{ij}$ for $i, j = 1, \ldots, k$.

For a $(k+1)$-dimensional vector space V, there exists a non-zero vector e_{k+1} such that $g(e_{k+1}, e_{k+1}) \neq 0$. If not, then $g(v, v) = 0$ for all $v \in V$, and

$$g(v, w) = \frac{1}{4}\left(g(v + w, v + w) - g(v - w, v - w)\right) = 0 \tag{46}$$

Which contradicts the non-degeneracy of g. Without loss of generality, we can assume that $g(e_{k+1}, e_{k+1}) = \pm 1$. Then, consider following two subspaces of V:

$$W = \{w \in V : g(e_{k+1}, w) = 0\}, \tag{47}$$

$$U = \mathrm{span}\{e_{k+1}\} \tag{48}$$

Then, clearly, $W \cap U = \{0\}$. Also, for any $v \in V$,

$$v = \underbrace{v \mp g(e_{k+1}, v)e_{k+1}}_{\in W} \pm \underbrace{g(e_{k+1}, v)e_{k+1}}_{\in U} \tag{49}$$

with same choice of sign as $g(e_{k+1}, e_{k+1}) = \pm 1$. Hence, $V = W \oplus U$ and $\dim W = \dim V - 1 = k$. Moreover, $g|_W$ is non-degenerate: if $w \in W$ satisfies $g(w, w') = 0$ for all $w' \in W$, then since also $g(w, e_{k+1}) = 0$ and $V = W \oplus U$, we have $g(w, v) = 0$ for all $v \in V$, hence $w = 0$. Thus $g|_W$ is a metric on W, and by the induction hypothesis, take a basis $\{e_1, \ldots, e_k\}$ of W such that $g(e_i, e_j) = \pm\delta_{ij}$ for $i, j = 1, \ldots, k$. By the definition of W, we have $g(e_i, e_{k+1}) = 0$ for $i = 1, \ldots, k$. Hence, $\{e_1, \ldots, e_k, e_{k+1}\}$ is a basis of V such that $g(e_i, e_j) = \pm\delta_{ij}$ for $i, j = 1, \ldots, k + 1$.

(b)

Let $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_n\}$ be two orthonormal bases of (V, g). By permuting the order if necessary, we can assume that

$$g(v_i, v_i) = \begin{cases} 1 & i = 1, \ldots, p \\ -1 & i = p + 1, \ldots, n \end{cases} \tag{50}$$

and

$$g(w_i, w_i) = \begin{cases} 1 & i = 1, \ldots, q \\ -1 & i = q + 1, \ldots, n \end{cases} \tag{51}$$

It is enough to show that $p = q$.

Define four subspaces of V as

$$V_+ = \mathrm{span}\{v_1, \ldots, v_p\}, \qquad V_- = \mathrm{span}\{v_{p+1}, \ldots, v_n\}, \tag{52}$$

$$W_+ = \mathrm{span}\{w_1, \ldots, w_q\}, \qquad W_- = \mathrm{span}\{w_{q+1}, \ldots, w_n\} \tag{53}$$

Then, g is positive definite on V_+ and W_+, and negative definite on V_- and W_-. Then, $V_+ \cap W_- = \{0\}$ because if $v \in V_+ \cap W_-$ is a non-zero vector, then $g(v,v) > 0$ and $g(v,v) < 0$, which is a contradiction. In the same way, $V_- \cap W_+ = \{0\}$. Hence, by the dimension formula ($\dim(A+B) = \dim A + \dim B - \dim(A \cap B)$),

$$\dim(V_+ + W_-) = \dim V_+ + \dim W_- = p + (n-q) \leq n \tag{54}$$

$$\dim(V_- + W_+) = \dim V_- + \dim W_+ = (n-p) + q \leq n \tag{55}$$

The two inequalities imply that $p \leq q$ and $q \leq p$. Hence, $p = q$.

#4

(a)

For a point p in $\mathbb{R}^3$ not on the z-axis has two coordinate neighborhoods (U, id), $(U, (r, \theta, \phi))$ with transition map

$$\begin{aligned} x &= r\sin\theta\cos\phi \\ y &= r\sin\theta\sin\phi \\ z &= r\cos\theta \end{aligned} \tag{56}$$

Their derivatives are

$$\begin{aligned} &\frac{\partial x}{\partial r} = \sin\theta\cos\phi, && \frac{\partial x}{\partial \theta} = r\cos\theta\cos\phi, && \frac{\partial x}{\partial \phi} = -r\sin\theta\sin\phi, \\ &\frac{\partial y}{\partial r} = \sin\theta\sin\phi, && \frac{\partial y}{\partial \theta} = r\cos\theta\sin\phi, && \frac{\partial y}{\partial \phi} = r\sin\theta\cos\phi, \\ &\frac{\partial z}{\partial r} = \cos\theta, && \frac{\partial z}{\partial \theta} = -r\sin\theta, && \frac{\partial z}{\partial \phi} = 0. \end{aligned} \tag{57}$$

Then, the metric, as a $(0,2)$-tensor, transforms as

$$\begin{aligned} ds^2 &= dx^2 + dy^2 + dz^2 \\ &= \left(\frac{\partial x}{\partial r}dr + \frac{\partial x}{\partial \theta}d\theta + \frac{\partial x}{\partial \phi}d\phi\right)^2 + \left(\frac{\partial y}{\partial r}dr + \frac{\partial y}{\partial \theta}d\theta + \frac{\partial y}{\partial \phi}d\phi\right)^2 + \left(\frac{\partial z}{\partial r}dr + \frac{\partial z}{\partial \theta}d\theta + \frac{\partial z}{\partial \phi}d\phi\right)^2 \\ &= dr^2 + r^2 d\theta^2 + r^2\sin^2\theta d\phi^2 \end{aligned} \tag{58}$$

In details, each coefficients are calculated as follows:

1. Coefficient of dr^2:
$$\left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2 = (\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta) = 1 \tag{59}$$

2. Coefficient of $d\theta^2$:
$$\left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2 = (r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta) = r^2 \tag{60}$$

3. Coefficient of $d\phi^2$:
$$\left(\frac{\partial x}{\partial \phi}\right)^2 + \left(\frac{\partial y}{\partial \phi}\right)^2 + \left(\frac{\partial z}{\partial \phi}\right)^2 = (r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi) = r^2\sin^2\theta \tag{61}$$

4. Coefficient of $drd\theta$:
$$2\frac{\partial x}{\partial r}\frac{\partial x}{\partial \theta} + 2\frac{\partial y}{\partial r}\frac{\partial y}{\partial \theta} + 2\frac{\partial z}{\partial r}\frac{\partial z}{\partial \theta} = 2(r\sin\theta\cos\theta\cos^2\phi + r\sin\theta\cos\theta\sin^2\phi - r\sin\theta\cos\theta) = 0 \tag{62}$$

5. Coefficient of $drd\phi$:
$$2\frac{\partial x}{\partial r}\frac{\partial x}{\partial \phi} + 2\frac{\partial y}{\partial r}\frac{\partial y}{\partial \phi} + 2\frac{\partial z}{\partial r}\frac{\partial z}{\partial \phi} = 2(-r\sin^2\theta\sin\phi\cos\phi + r\sin^2\theta\cos\phi\sin\phi) = 0 \tag{63}$$

6. Coefficient of $d\theta d\phi$:
$$2\frac{\partial x}{\partial \theta}\frac{\partial x}{\partial \phi} + 2\frac{\partial y}{\partial \theta}\frac{\partial y}{\partial \phi} + 2\frac{\partial z}{\partial \theta}\frac{\partial z}{\partial \phi} = 2(-r^2\cos\theta\sin\theta\cos\phi\sin\phi + r^2\cos\theta\sin\theta\sin\phi\cos\phi) = 0 \tag{64}$$

(b)

Consider the flat spacetime $\mathbb{R}^{1,3}$ with two chart $(\mathbb{R}^{1,3}, \mathrm{id})$ and $(\mathbb{R}^{1,3}, \varphi)$ where $\varphi = (t', x', y', z')$ is rotating coordinate system defined by the transition map

$$\begin{aligned} t' &= t \\ x' &= (x^2 + y^2)^{1/2} \cos(\phi - \omega t) \\ y' &= (x^2 + y^2)^{1/2} \sin(\phi - \omega t) \\ z' &= z \end{aligned} \tag{65}$$

Where ϕ is the azimuthal angle in cylindrical coordinates.

Using $\cos\phi = x/(x^2+y^2)^{1/2}$ and $\sin\phi = y/(x^2+y^2)^{1/2}$, we can rewrite the transition map as

$$\begin{aligned} t' &= t \\ x' &= x\cos(\omega t) + y\sin(\omega t) \\ y' &= -x\sin(\omega t) + y\cos(\omega t) \\ z' &= z \end{aligned} \tag{66}$$

and the inverse is

$$\begin{aligned} t &= t' \\ x &= x'\cos(\omega t') - y'\sin(\omega t') \\ y &= x'\sin(\omega t') + y'\cos(\omega t') \\ z &= z' \end{aligned} \tag{67}$$

Their derivatives are

$$\begin{aligned} &\frac{\partial t}{\partial t'} = 1, && \frac{\partial t}{\partial x'} = 0, && \frac{\partial t}{\partial y'} = 0, && \frac{\partial t}{\partial z'} = 0, \\ &\frac{\partial x}{\partial t'} = -\omega x'\sin(\omega t') - \omega y'\cos(\omega t'), && \frac{\partial x}{\partial x'} = \cos(\omega t'), && \frac{\partial x}{\partial y'} = -\sin(\omega t'), && \frac{\partial x}{\partial z'} = 0, \\ &\frac{\partial y}{\partial t'} = \omega x'\cos(\omega t') - \omega y'\sin(\omega t'), && \frac{\partial y}{\partial x'} = \sin(\omega t'), && \frac{\partial y}{\partial y'} = \cos(\omega t'), && \frac{\partial y}{\partial z'} = 0, \\ &\frac{\partial z}{\partial t'} = 0, && \frac{\partial z}{\partial x'} = 0, && \frac{\partial z}{\partial y'} = 0, && \frac{\partial z}{\partial z'} = 1. \end{aligned} \tag{68}$$

Hence, the metric, transforms as

$$\begin{aligned} ds^2 &= -dt^2 + dx^2 + dy^2 + dz^2 \\ &= -\left(\frac{\partial t}{\partial t'}dt' + \frac{\partial t}{\partial x'}dx' + \frac{\partial t}{\partial y'}dy' + \frac{\partial t}{\partial z'}dz'\right)^2 + \left(\frac{\partial x}{\partial t'}dt' + \frac{\partial x}{\partial x'}dx' + \frac{\partial x}{\partial y'}dy' + \frac{\partial x}{\partial z'}dz'\right)^2 \\ &+ \left(\frac{\partial y}{\partial t'}dt' + \frac{\partial y}{\partial x'}dx' + \frac{\partial y}{\partial y'}dy' + \frac{\partial y}{\partial z'}dz'\right)^2 + \left(\frac{\partial z}{\partial t'}dt' + \frac{\partial z}{\partial x'}dx' + \frac{\partial z}{\partial y'}dy' + \frac{\partial z}{\partial z'}dz'\right)^2 \\ &= -\left(1 - \omega^2(x'^2 + y'^2)\right)dt'^2 + 2\omega(x'dy' - y'dx')dt' + dx'^2 + dy'^2 + dz'^2 \end{aligned} \tag{69}$$

And the coefficients are

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \omega^2(x'^2 + y'^2)\right) & -\omega y' & \omega x' & 0 \\ -\omega y' & 1 & 0 & 0 \\ \omega x' & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{70}$$

For the inverse metric, we can use the formula for the inverse of a block matrix:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} (A - BD^{-1}C)^{-1} & -(A - BD^{-1}C)^{-1}BD^{-1} \\ -D^{-1}C(A - BD^{-1}C)^{-1} & D^{-1} + D^{-1}C(A - BD^{-1}C)^{-1}BD^{-1} \end{pmatrix} \tag{71}$$

In our case, $D = D^{-1} = I$. Hence, we have

$$g^{\mu\nu} = \begin{pmatrix} -1 & -\omega y' & \omega x' & 0 \\ -\omega y' & 1 - \omega^2 y'^2 & \omega^2 x' y' & 0 \\ \omega x' & \omega^2 x' y' & 1 - \omega^2 x'^2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{72}$$